

A Comparative Evaluation of Generative Adversarial Networks for Channel Modeling in Coded MIMO-OFDM Systems

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Abstract—In 5G and beyond communication, the combination of deep learning (DL) and digital twin (DT) channel model makes the system improve efficiency. By using the generative model of DL, the specified statistical distribution, the channel feature as channel impulse response is generated from the measurement data. This paper compares the schemes of Generative adversarial networks (GAN), Variational Auto-encoder GAN (VAE-GAN) and Wasserstein GAN with gradient penalty (WGAN-GP) in acquiring the BERs of SISO/MIMO system with different delays from the real data. From the simulation results, the channel characters are re-edified exactly referred to as the original channel, especially in terms of WGAN-GP. Moreover, of the LDPC decoding algorithms as Linear Approximation Min-Sum (LAMS) and Belief Propagation (BP), the performance of LAMS-LDPC is the best in case of employing WGAN-GP in MIMO-OFDM system. If the SNR is increased ($\text{SNR} > 20\text{dB}$), the better BERs of the WGAN-GP are. Therefore, we propose to exploit LDPC-LAMS combined to WGAN-GP in our coded MIMO-OFDM system.

Index Terms—LDPC, MIMO-OFDM, 5G channel modeling, Generative Adversarial Network, GAN, WGAN-GP, VAE-GAN.

I. INTRODUCTION

Multiple-Input Multiple-Output (MIMO) combined with Orthogonal Frequency-Division Multiplexing (OFDM) has become a fundamental technology in modern broadband wireless communication systems. By transmitting multiple data streams across several antennas and orthogonal subcarriers, MIMO-OFDM systems can achieve high spectral efficiency and strong robustness against multipath fading [1]. To further enhance reliability, LDPC codes have been widely adopted in wireless standards such as IEEE 802.11n, WiMAX, and 5G New Radio (NR) due to their near-Shannon limit error-correction capability [2]. Among LDPC decoding algorithms [3] as 3GPP standard, the BP algorithm is considered the most

accurate approach, as it performs iterative message passing between variable and check nodes to achieve optimal error correction. However, this accuracy comes at the cost of high computational complexity and decoding latency, which make BP less suitable for real-time or hardware-constrained systems. To overcome these limitations, simplified algorithms such as the LAMS method have been proposed [4]- [7]. LAMS modifies the message-passing process to reduce computational overhead and enable faster convergence while maintaining BER performance close to that of BP decoding. Therefore, analyzing the trade-off between decoding complexity and error-rate performance of BP and LAMS is a key step in designing efficient LDPC-coded systems.

The DL techniques in advanced 5G/6G are trending with the real data distribution in the case of how to deal with distributions supported by low dimensional manifolds. As consequences, the virtually generative models illustrate a system with noise in high bandwidth are transpires as the solution in wireless communication channel. Accurate channel modeling is another crucial factor that influences the evaluation of the performance in MIMO-OFDM systems. Conventional analytical models such as Rayleigh or Rician fading describe statistical channel behavior but fail to capture the spatial, temporal, and non-linear dynamics of real-world wireless propagation. To address this limitation, data-driven modeling techniques have emerged, particularly GANs, which can learn complex distributions from measured data and generate synthetic channel datasets that preserve real-world characteristics [9]. GAN-based channel modeling has shown potential in reproducing key statistical properties such as path loss, shadow fading, and channel impulse response (CIR), enabling more realistic and scalable simulations [10]- [15]. To deploy the GAN techniques with huge training data process, the DT [16] can function as virtual environment that replicating the physical channel

then facilitate flexible simulation and performance analysis of wireless communication systems. From GAN, the SISO system performance [17] is obtained after training measured data in virtual layer. For describing the scenarios of line-of-sight (LOS) propagation channel from WINNER II as 3GPP standard for pedestrian, vehicular and hilly terrain, as well as and the SUI channel types, authors [18] investigated the SISO channel characteristic.

In this work, three generative models—standard GAN, WGAN-GP, and VAE-GAN are used to synthesize channel data that mimic real measured environments. Both real and synthetic datasets are then utilized within a complete LDPC-coded MIMO-OFDM simulation. The evaluation consists of two main analyses. First, a data-level comparison assesses how accurately each generative model reproduces the measured channel characteristics through statistical metrics such as CIR variation, path loss probability density, and shadow fading distribution. The measured and generated results are illustrated in the probability density function (PDF) and cumulative distribution function (CDF) plots. Second, a performance-level comparison investigates the decoding behavior of BP and LAMS algorithms under both measured and synthetic channel datasets. The BER graphs reveal how decoding performances change when substituting real data with GAN-based data.

The simulation results show that GAN-generated data, particularly from the WGAN-GP model, achieve high fidelity with the measured dataset, producing BER curves that closely follow the real-data baseline. Moreover, while the BP decoder consistently achieves the lowest BER, the LAMS algorithm demonstrates a highly favorable trade-off between decoding accuracy and computational efficiency, achieving similar BER performance at significantly lower decoding complexity. These findings confirm that GAN-based channel data can be effectively used for performance evaluation, reducing the dependency on extensive measurement campaigns while maintaining the reliability of simulation results. Our paper is structured as followings. The channel decoding LDPC algorithms are present in Part II, The GAN-based are analyzed in Part III. The Simulation Results in Code MIMO-OFDM are given in Part IV, then the conclusions are mentioned in the final Part.

II. CHANNEL CODING OF LPDC

A. LDPC Decoder Algorithm

The 3GPP channel coding standards for 5G [2] are classified two structured base graphs for long information blocks and low code (BG1) or shorter blocks and higher rates (BG2) to illustrate the sparse matrix $\mathbf{H}$. LDPC decoding algorithms are considered as the BP and LAMS algorithm [7]. Both iteratively exchange messages between variable nodes (VNs) and check nodes (CNs) until convergence or reaching the iteration limit.

1) *Belief Propagation (BP)-based Algorithm:* In the BP decoder, the message from check node y to variable node x is updated as:

$$L_{y \rightarrow x} = 2 \tanh^{-1} \left(\prod_{x' \in B(y) \setminus x} \tanh \frac{L_{x' \rightarrow y}}{2} \right) \quad (1)$$

where $B(y)$ is the set of variable nodes connected to check node y . The final log-likelihood ratio (LLR) is attained by the loop of initial LLR and the loop of updated message at each iteration. While BP provides high decoding accuracy, it requires nonlinear $\tanh$ operations, making it costly for hardware implementation.

2) *Linear Approximation Minimum-Sum (LAMS) Algorithm:* The LAMS decoder simplifies BP by using linear approximations and two correction factors to reduce computation and maintain accuracy [16]. The message from check node y to variable node x is given by:

$$L_{y \rightarrow x} = \prod_{x' \in B(y) \setminus x} \text{sign}(L_{x' \rightarrow y}) \times \max \left(\alpha \times \min_{x' \in B(y) \setminus x} |L_{x' \rightarrow y}| + \beta, 0 \right) \quad (2)$$

where α is a normalization factor ($0 < \alpha \leq 1$) and β is an offset parameter. These adjustments reduce the overestimation error and remove dependence on the noise variance, allowing LAMS to achieve near-BP performance with significantly lower complexity—well suited for real-time MIMO systems.

III. GENERATIVE ADVERSARIAL NETWORK FOR WIRELESS CHANNEL MODELING

A. Generative Adversarial Network

The GAN model [9], [16] - [18] has recently been employed in wireless communications to model and regenerate complex channel characteristics to capture the conventional statistical models. In a MIMO-OFDM system, the CIR can be represented as a three-dimensional tensor of: $h \in \mathbb{C}^{N_r \times N_t \times D}$; where N_r and N_t is the number of receive and antennas, respectively; D denotes the number of delay taps. The goal of the GAN is to learn the empirical distribution $p_{data}(h)$ of real CIR samples and generate synthetic channel responses $\hat{h}$ that share similar spatial-temporal characteristics.

The GAN consists of two neural networks: a *generator* G and a *discriminator* D . The generator $G(z; \theta_G)$ maps a random latent vector $z \sim p_z(z)$, typically Gaussian noise, into a synthetic CIR sample $\hat{h} = G(z)$. The discriminator $D(h; \theta_D)$ receives either real or generated CIR samples and outputs the likelihood that the sample originates from real measurements. Both networks are trained in a minimax game defined by:

$$\min_G \max_D V(D, G) = \mathbb{E}_{h \sim p_{data}(h)} [\log D(h)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))] \quad (3)$$

The generator aims to produce CIRs that the discriminator cannot distinguish from real ones, while the discriminator learns to maximize its accuracy. At convergence, the generated distribution $p_g(h)$ approximates the real distribution $p_{data}(h)$, allowing realistic modeling of MIMO propagation channels.

B. Wasserstein GAN with Gradient Penalty (WGAN-GP)

To address training instability and mode collapse in conventional GANs, the WGAN [10] - [12] replaces the Jensen-Shannon divergence with the Wasserstein-1 (Earth Mover's) distance, providing smoother gradients during optimization. The WGAN objective is expressed as:

$$L = \mathbb{E}_{h \sim p_{data}(h)}[D(h)] - \mathbb{E}_{z \sim p_z(z)}[D(G(z))], \quad (4)$$

where D (also referred to as a *critic*) outputs real-valued scores instead of probabilities and must satisfy the 1-Lipschitz constraint. To enforce this constraint, the WGAN with Gradient Penalty (WGAN-GP) [16] adds a regularization term:

$$L_{GP} = L + \lambda \mathbb{E}_{\hat{h} \sim p_{\hat{h}}} \left[(\|\nabla_{\hat{h}} D(\hat{h})\|_2 - 1)^2 \right] \quad (5)$$

where $\hat{h}$ denotes samples linearly interpolated between real and generated CIRs, and λ is a penalty coefficient. This method stabilizes the learning process and enables the model to capture realistic spatial-temporal dependencies in MIMO fading channels.

C. Variational Autoencoder GAN (VAE-GAN)

To enhance the stability and generalization capability of GAN-based channel models, the VAE-GAN [13]- [15] combines the latent representation learning of Variational Autoencoders (VAE) with the adversarial loss of GANs. The VAE optimizes the evidence lower bound (ELBO) as following:

$$L_{VAE} = \mathbb{E}_{q(z|h)}[\log p(h|z)] - D_{KL}(q(z|h)||p(z)), \quad (6)$$

where $q(z|h)$ is the encoder that maps the real CIR to a latent variable z , $p(h|z)$ is the decoder reconstructing the CIR, and D_{KL} denotes the Kullback-Leibler divergence. In the VAE-GAN framework, the decoder functions as the generator G in the adversarial setting, with an additional loss component:

$$L_{VAE-GAN} = L_{VAE} + \lambda_{adv} \left(\mathbb{E}_{h \sim p_{data}}[\log D(h)] + \mathbb{E}_{z \sim p_z}[\log(1 - D(G(z)))] \right) \quad (7)$$

where λ_{adv} balances the reconstruction and adversarial terms. This hybrid architecture leverages the robust encoding ability of the VAE, and further improves the realism of the generated CIRs.

IV. SIMULATION AND RESULTS

In this section, we describe the real measurement dataset and the training configuration used for all above GAN-based models, followed by an evaluation of their effectiveness compared to experimentally measured CIRs reported in [3]. The measurement campaign was conducted over the 55–65 GHz band using a vector network analyzer and open-ended waveguide antennas in indoor environments in both LOS and NLOS conditions. A total 160 channel transfer function (CTF) snapshots were captured. Each CTF is converted into a CIR using the IEEE 802.11ad channel estimation field, resulting in 128 delay-domain samples per CIR. To focus on the dominant multipath components and to match the 802.11ad

signal bandwidth, each CIR is down-sampled and only the first 32 delay taps are retained. These 160 CIR realizations are used as the real data distribution for training the GAN-based models. The generated CIR sets are then fed into a 4×4 channel-coded MIMO-OFDM communication system implemented in MATLAB to examine the BER performance.

A. Training the GAN-based Models

The GAN, WGAN-GP, and VAE-GAN models are implemented in PyTorch and trained using the hyperparameters summarized in Table I. The loss functions can be estimated for GAN and WGAN-GP as: Binary cross-entropy loss and Wasserstein loss with added gradient penalty, respectively, while loss function for VAE-GAN is a combination of Kullback-Leibler Divergence (KLD) loss for the latent distribution and Wasserstein loss.

TABLE I. Training Hyperparameters for SISO Models

Parameter	Value
Optimizer	Adam
Learning Rate	5×10^{-5}
β_1	0.5
β_2	0.9
Batch Size	64
Gradient Penalty Coefficient (λ)	10 (WGAN-GP, VAE-GAN)
Critic Update Frequency	5 (per 1 Generator update)
Training Epochs	5000

Figures 1, 2, 3, and 4 respectively show the Power Delay Profile (PDP) of the measured channel, the GAN-generated channel, the WGAN-GP-generated channel, and the VAE-GAN-generated channel. However, instead of plotting the delay on the vertical axis, the figures represent the corresponding transmission path distances. It can be observed that the generated channels appear quite realistic and closely resemble the measured data. The PDPs generated by those GAN-based methods show a good level of similarity to the real measured data. Specifically, by examining the major power-contributing paths marked with numbered annotations and comparing them with Figure 1, we observe as the followings:

- 1: At a distance of 2.5 meters, this is the strongest power-contributing path, corresponding to the direct line-of-sight (LOS) between the transmitter and receiver.
- 2, 3: Located at 2.6 meters and 3 meters, these correspond to reflected paths from metallic table legs.
- 4: At a distance of 3.3 meters, corresponding to ground reflections.
- 5, 6: Around 4 meters, these paths are due to reflections from a whiteboard.
- 7: At 4.8 meters, despite the relatively long distance, a strong reflection is observed due to metallic lighting frames.

After applying the Fourier transform to the CIR datasets, we obtain Figure 5, 6, 7, and 8, which illustrates the signal power of the CTF versus frequency. Similar to the PDP analysis, the three generated channels also exhibit a high level of similarity to the experimentally measured channel. This is reflected in the

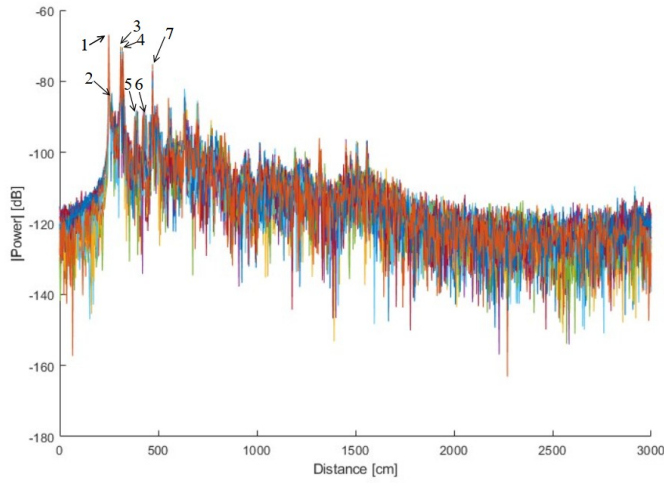


Fig. 1. PDP of the measured wireless channel.

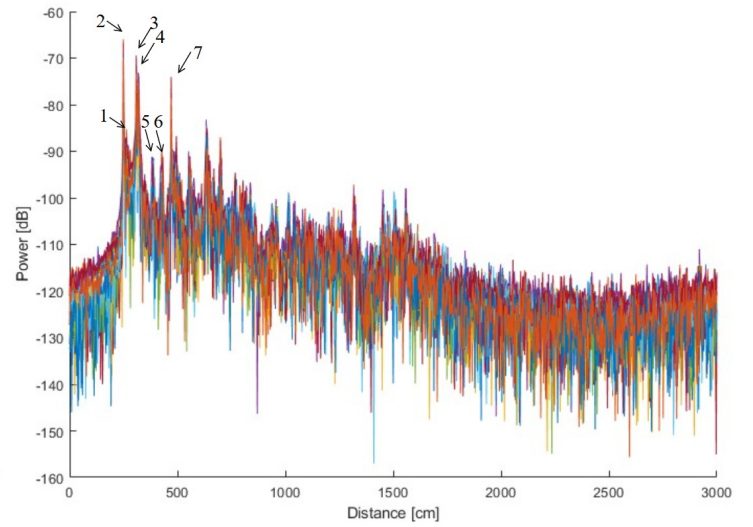


Fig. 4. PDP of the channel generated by VAE-GAN.

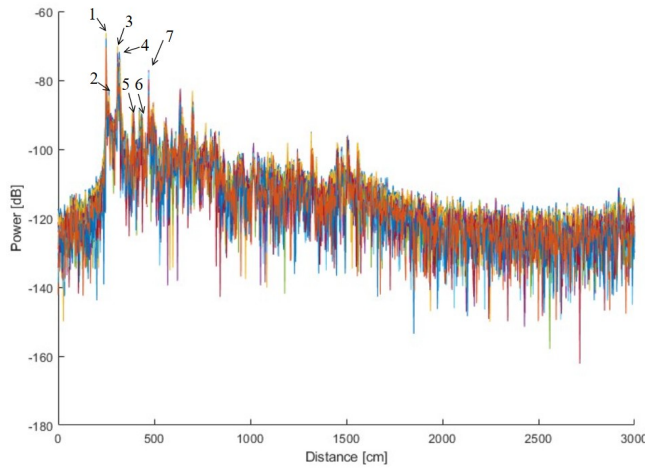


Fig. 2. PDP of the channel generated by GAN.

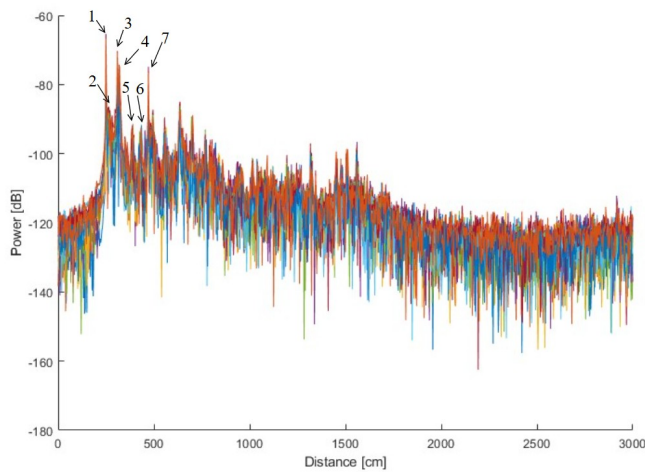


Fig. 3. PDP of the channel generated by WGAN-GP.

signal power fluctuations, which mainly lie within the range of -60 dB to -160 dB. Furthermore, it can be that all four channels exhibit frequency selectivity with significant variations across the frequency domain.

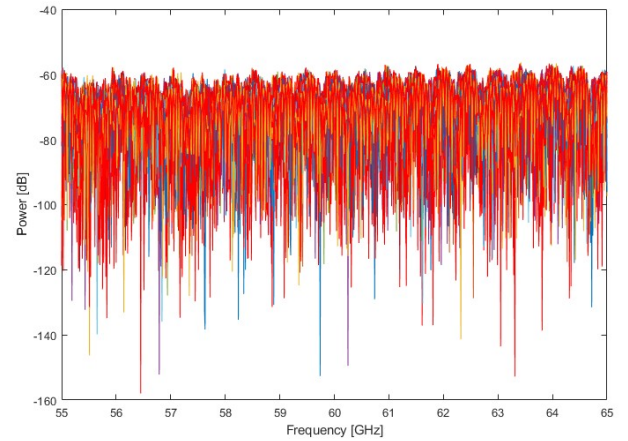


Fig. 5. CTF Signal Power versus Frequency for the Measured Channel.

Figures 9 and 10 illustrate the PDF and CDF of shadow fading, respectively. The PDF in Fig. 9, there are similar shape in all generative models to the measured one. The CDF plot in Fig. 10, the curves almost completely overlap, indicating good training performance across all three models. It can be observed that all four datasets follow a Gaussian distribution with a mean of zero. The standard deviations for the Measured Channel, GAN- Channel, WGAN-GP- Channel, and VAE-GAN Channel are 10,9450; 11,0187; 10,9141; and 10,9059, respectively. Among them, the WGAN-GP channel yields the closest standard deviation to the real measurements. However,

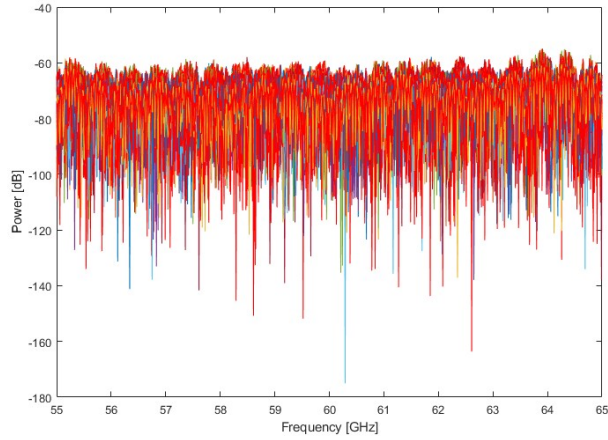


Fig. 6. CTF Signal Power versus Frequency for the DT GAN Channel.

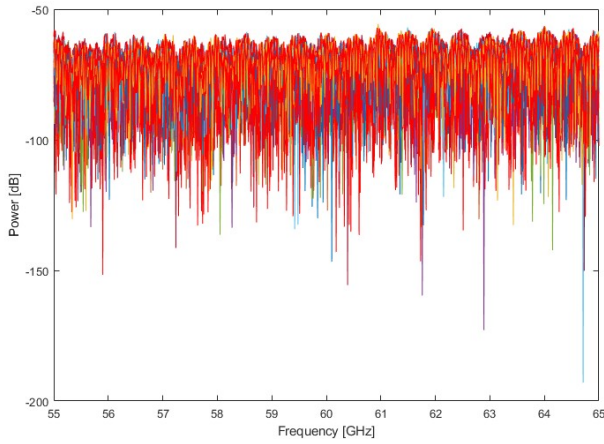


Fig. 7. CTF Signal Power versus Frequency for the DT WGAN-GP Channel.

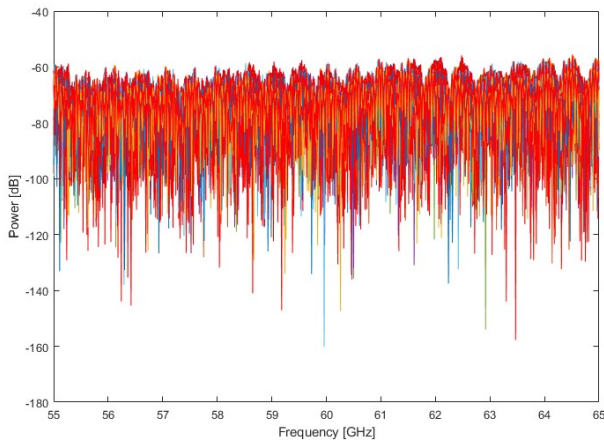


Fig. 8. CTF Signal Power versus Frequency for the DT VAE-GAN Channel.

overall, the differences in standard deviation among the three GAN-based models are negligible.

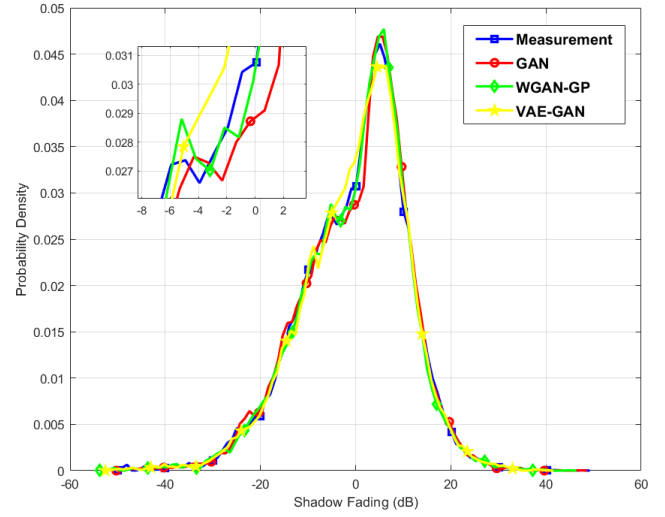


Fig. 9. Probability Density Function (PDF) of shadow fading.

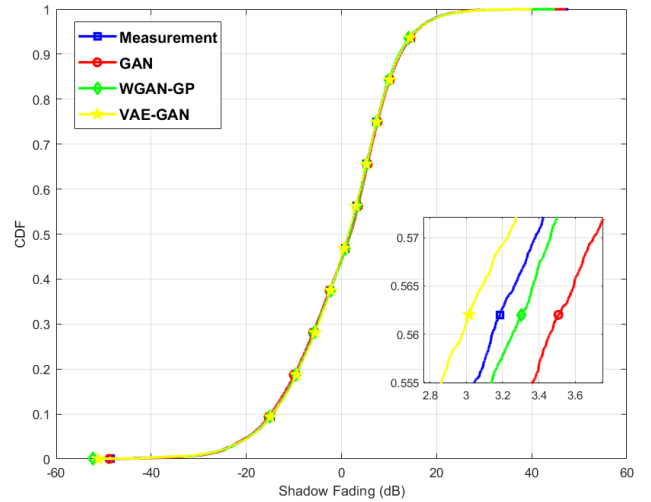


Fig. 10. The CDF of shadow fading.

Similar to shadow fading, the PDF and CDF of path loss are illustrated in Figure 11 and 12, respectively. However, it is easier to observe the discrepancies between the GAN and VAE-GAN-generated channels compared to the measured data, as the red line with circular markers and the yellow line with star-shaped markers do not align well with the blue line with square markers. In contrast, the WGAN-GP-generated channel shows strong agreement with the blue line in both figures. This observation is further supported by the statistical results: the mean path loss values for the measured, GAN, WGAN-GP, and VAE-GAN channels are 115,4371; 117,5786;

115,0317; and 117,0161, respectively. It can be concluded that, among the three models, WGAN-GP yields the most accurate approximation of real-world path loss. Nonetheless, the GAN and VAE-GAN models also produce reasonably good results, and the differences are not significantly large, making them worth further consideration.

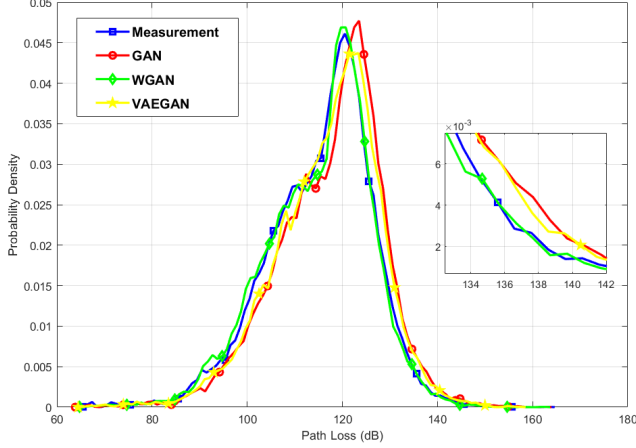


Fig. 11. The PDF of pathloss.

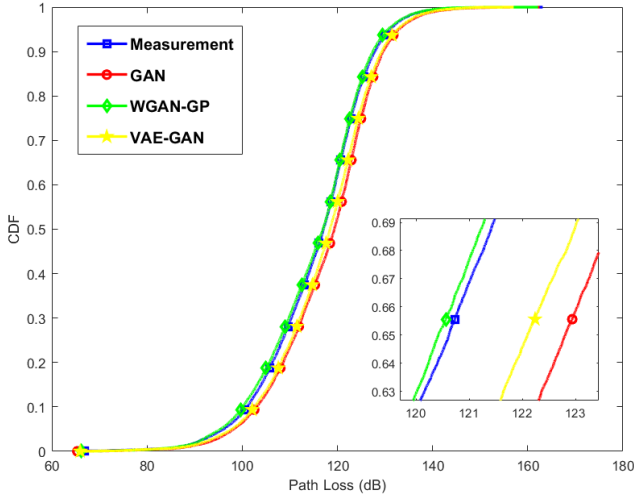


Fig. 12. The CDF of pathloss.

B. Simulation and Performance Evaluation of 4×4 MIMO-OFDM Communication System

In this section, a MIMO-OFDM system illustrated in Figure 13 is simulated using both the experimentally measured channel and the three channels generated by the GAN-based models. LDPC channel coding is employed to enhance the reliability of the input bitstream, while 64QAM modulation is used to achieve a higher bit rate compared to other modulation

schemes. Space-Frequency Block Coding (SFBC) is integrated with the OFDM technique. The receiver side do inversly as the transmitter except the channel equalization methods—ZF and MMSE to recover the signal. The system parameters are summarized in Table II.

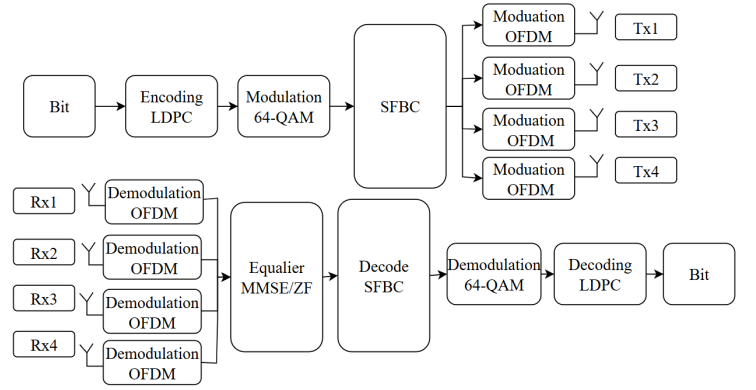


Fig. 13. Block diagram of the MIMO-OFDM system.

TABLE II. Simulation parameters

Parameter	Value
Environment	Indoor
Number of subcarriers	512
Number of OFDM symbols	14
Number of antennas	MIMO 4×4 / SISO
Modulation	64-QAM
Code rate	1/3
LDPC decoder	BP / LAMS
BP decoding algorithm	SCL
LDPC iterations	3

The selected channel data corresponds to the transmission from antenna 1 at the transmitter side to antenna 1 at the receiver side. This choice is made because the overall number of available channels is too large to present comprehensively in this report, and the channels exhibit similar characteristics, leading to consistent results across different links. In Figure 14, in SISO system, when comparing the GAN-based models with the experimentally measured channel, it is evident that the graph of WGAN-GP shows the highest similarity to the measured channel. Although the GAN and VAE-GAN models are further away from the measured channel graph, they still exhibit a high level of similarity. Additionally, to evaluate the most suitable channel decoding method for the system, both the BP and LAMS algorithms are applied to the LDPC decoder. As one can be observed, it is found that at low SNR values (<16 dB), BP achieves lower BER than LAMS. The BER curves of BP decrease gradually if SNRs increase, whereas LAMS maintains a nearly flat BER at low SNRs. However, once the SNR exceeds a certain threshold (>16 dB), the BERs of LAMS drops significantly, making it become outperforming to BP's. Specifically, for a BER range

of approximately 10^{-3} to 10^{-4} , LAMS only requires an SNR of 18 dB, while BP requires up to 24 dB. The WGAN-GP is closer to real channel compared to other GAN models.

A similar trend can be attained in the case of the MIMO-OFDM 4×4 system, as shown in Figure 15. These results suggest that LAMS provides significantly better decoding performance than BP when the system operates in a sufficiently high SNR regime. Once again, the WGAN-GP graphs are next to the real channel while other models have a significant gap.

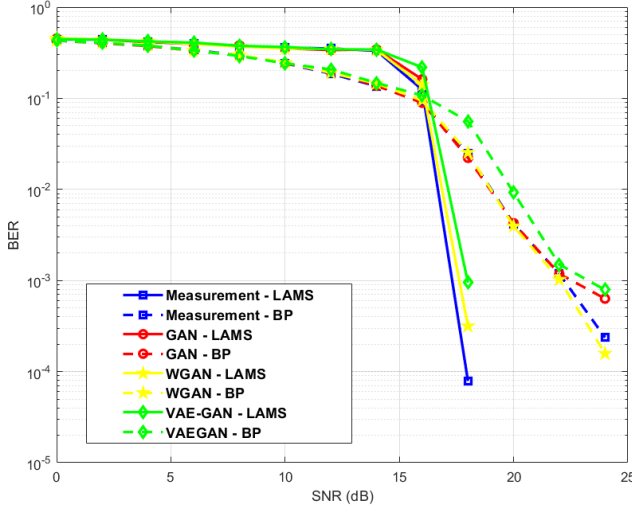


Fig. 14. BER performance of the system using a SISO antenna with BP and LAMS channel decoding techniques.

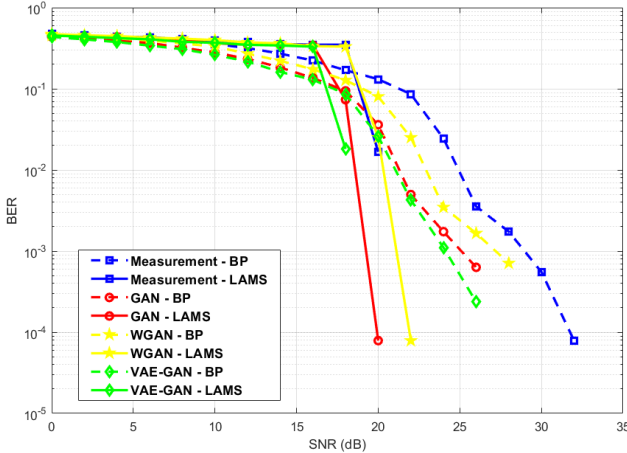


Fig. 15. BER performance of the system using a MIMO antenna with BP and LAMS channel decoding techniques.

Figure 16 shows the comparison of system performance using two channel equalization methods: ZF and MMSE. For all four channels, the BER of the MMSE equalizer is consistently lower than that of ZF. That is, MMSE considers the noise power during its computation and effectively reduces the noise

impact on the original signal. Of the three generative models, the WGAN-GP are capable of imitating the measured channel.

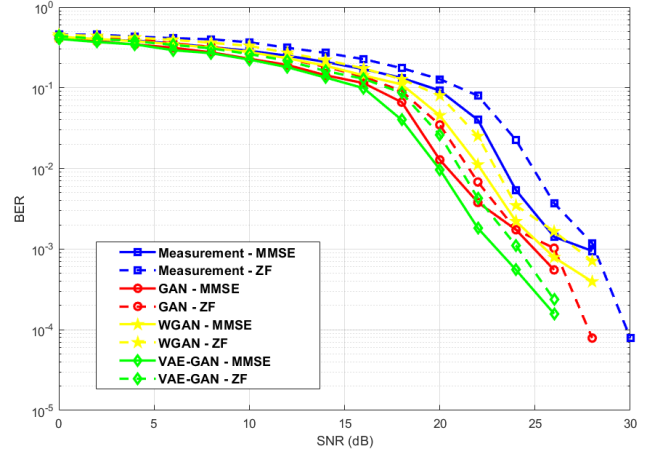


Fig. 16. BER performance of the system using a MIMO antenna with MMSE and ZF channel equalization technique.

V. CONCLUSION

In this study, we proposed and implemented a digital twin channel modeling approach based on deep generative networks, including GAN, WGAN-GP, and VAE-GAN. The models were trained on real measured CIR data under SISO or 4×4 MIMO configurations, aiming to generate artificial channel data that closely match the statistical properties of real-world measurements. Our experimental results demonstrate that WGAN-GP outperforms to other models in replicating critical channel characteristics such as the Power Delay Profile (PDP), Path Loss, and Shadow Fading. Evaluations using distribution plots, as CDF, PDF and system-level BER simulations confirm that the WGAN-GP generated data exhibits the highest similarity to real data. Among the GAN-based schemes, we propose to use the WGAN-GP due to the most effective in our MIMO-OFDM using LDPC Channel coding. In our BER simulation results, the LAMS decoding is more effective than the BP's.

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REFERENCES

- [1] L. Pang, J. Zhang, et al., "Investigation and comparison of 5g channel models: From quadriga, nysim, and mg5g per spectives," Chinese Journal of Electronics, vol. 31, no. 1, pp. 1–17, 2022.
- [2] 3GPP TS 38.212, Multiplexing and channel coding, V15.4.0, Dec 2018.
- [3] J. Blumenstein, J. Milos, L. Polak and C. Mecklenbräuker, "IEEE 802.11ad SC-PHY Layer Simulator: Performance in Real-world 60 GHz Indoor Channels," 2019 IEEE Nordic Circuits and Systems Conference (NORCAS): NORCHIP and International Symposium of System-on-Chip (SoC), Helsinki, Finland, 2019, pp. 1–4, doi: 10.1109/NORCHIP.2019.8906960.

- [4] N. T. Nga, N. Van Huy, N. M. Phuc, N. H. Hung and L. T. Thai, "A Comparison of LDPC and Polar Code on High Level Modulated MIMO-OFDM-EAC-NOMA System," 2025 19th International Conference on Ubiquitous Information Management and Communication (IMCOM), Bangkok, Thailand, 2025, pp. 1-7, doi: 10.1109/IMCOM64595.2025.10857515.
- [5] M. C. Davey and D. MacKay, "Low-density parity check codes over GF(q)," in *IEEE Communications Letters*, vol. 2, no. 6, pp. 165-167, June 1998, doi: 10.1109/4234.681360.
- [6] D. S. Shafiullah, M. R. Islam, Mohammad Mostafa Amir Faisal and I. Rahman, "Optimized Min-Sum decoding algorithm for Low Density PC codes," 2012 14th International Conference on Advanced Communication Technology (ICACT), PyeongChang, Korea (South), 2012, pp. 475-480.
- [7] M. J. X. Wu and C. Zhao, "Learning, Decoding Optimization for 5G LDPC Codes by Machine," *IEEE Access*, Vol. 6, pp. 50179 - 50186, 2018.
- [8] Y. Yang et al., "DECCO: Deep-Learning Enabled Coverage and Capacity Optimization for Massive MIMO Systems," *IEEE Access*, vol. 6, no. 1, Dec. 2018, pp. 23,361–71.
- [9] I. Goodfellow, et al., "Generative adversarial networks," *Communications of the ACM*, vol. 63, no. 11, pp. 139–144, 2020.
- [10] M. Bellemare, et al., "The Cramér Distance as a Solution to Biased Wasserstein Gradients," in *Proc. 35th Int. Conf. on Machine Learning (ICML)*, 2018.
- [11] M. Arjovsky, S. Chintala, et al., "Wasserstein generative adversarial networks," in *International Conference on Machine Learning*, pp. 214–223, 2017.
- [12] I. Gulrajani, et al., "Improved training of wasserstein gans," *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [13] Diederik P. Kingma; Max Welling, *An Introduction to Variational Autoencoders*, now, 2019.
- [14] S. Gur, S. Benaim, and L. Wolf, "Hierarchical Patch VAE-GAN: Generating Diverse Videos from a Single Sample," *arXiv preprint arXiv:2006.12226*, 2020. [Online]. Available: <https://arxiv.org/abs/2006.12226>
- [15] A. B. L. Larsen, S. K. Sønderby, H. Larochelle, and O. Winther, "Autoencoding beyond pixels using a learned similarity metric," in *Proc. of the 33rd International Conference on Machine Learning*, vol. 48, 2016, pp. 1558–1566.
- [16] Y. Yang, Y. Li, W. Zhang, F. Qin, P. Zhu and C. -X. Wang, "Generative-Adversarial-Network-Based Wireless Channel Modeling: Challenges and Opportunities," in *IEEE Communications Magazine*, vol. 57, no. 3, pp. 22-27, March 2019, doi: 10.1109/MCOM.2019.1800635.
- [17] Y. Zhang et al., "Generative adversarial networks based digital twin channel modeling for intelligent communication networks," in *China Communications*, vol. 20, no. 8, pp. 32-43, Aug. 2023, doi: 10.23919/JCC.fa.2023-0206.202308.
- [18] J. Wang, C. Wang, P. Wu and M. Xia, "Digital Twin Channel Modeling With Generative Adversarial Networks and Transfer Learning," 2025 *IEEE International Conference on Machine Learning for Communication and Networking (ICMLCN)*, Barcelona, Spain, 2025, pp. 1-6, doi: 10.1109/ICMLCN64995.2025.11139932.